Gurshaan Mand

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Education

University of Manitoba **Winnipeg, MB**
Bachelor of Science in Computer Science Expected May 2028
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming

Projects

Video Condenser **Python | FastAPI | Sentence Transformers | FFmpeg | Docker | AWS**

- Built an **objective-driven NLP video condensation platform** that reduced long-form YouTube video runtime by **up to 50%**, transforming time-aligned transcripts into focused, downloadable MP4 videos tailored to user-defined goals.
- Engineered a **multi-signal semantic ranking algorithm** using Sentence Transformers, 384-dimensional embeddings, cosine similarity, keyword relevance, topic coherence, novelty, and redundancy to identify high-value segments while preserving contextual continuity.
- Architected an asynchronous **FastAPI REST API** with Pydantic validation, background-job processing, progress polling, bounded concurrency, atomic job persistence, health checks, restart recovery, and FFmpeg-based H.264/AAC video rendering.
- Containerized and deployed the full-stack application through **Docker, Amazon ECR, and AWS ECS Express Mode**, configuring IAM access, Linux/AMD64 compatibility, offline model inference, logging, health monitoring, and public HTTPS delivery.

CareConnect — 24-Hour Hackathon Project (Team of 4) **Python | FastAPI | Pandas | Jinja2 | Priority Queues**

- Developed a **full-stack vaccination prioritization application** that processed 125 schedule records spanning 24 vaccine codes, generated synthetic patient profiles, and delivered patient-lookup and staff-dashboard workflows using FastAPI, Pandas, Faker, and Jinja2.
- Built an **explainable weighted-risk scoring engine** that evaluated age, vaccination gaps, and medical risk factors, then ranked patients by urgency using a max-heap priority queue while surfacing the reasons behind each score.

Allergen Finder **Python | SQL | SQLite | REST APIs | RapidFuzz**

- Reduced unnecessary Spoonacular API requests by building a **cache-first allergen detection pipeline** that queried SQLite, extracted allergens from ingredient metadata, and persisted unseen recipes through parameterized SQL.
- Improved search reliability for misspelled and partial food names by implementing **RapidFuzz-based category and recipe matching** with confidence thresholds, input normalization, duplicate detection, and REST API fallback.

Cave Explorer **Java | DFS | Linked Lists | Stacks**

- Implemented an **iterative depth-first search algorithm** to navigate a two-dimensional cave, detect obstacles, enforce boundaries, track visited cells, and locate a valid route to the treasure.
- Engineered a custom **linked-list-backed stack and explicit backtracking workflow** to recover from dead ends, prevent repeated traversal, and reconstruct the successful path through the cave.

Experience

Outlier AI **May 2025 – Aug 2025**
AI Data Contributor

- Supported **human-in-the-loop AI training** by evaluating LLM responses for factual accuracy, reasoning quality, instruction following, relevance, concision, and linguistic consistency.
- Identified hallucinations, reasoning failures, and inconsistent outputs, producing structured annotations and actionable feedback for model-evaluation datasets.

Walmart **April 2026 – Present**
Loss Prevention Officer

- Investigated suspicious activity using surveillance and transaction information, documenting findings while applying confidentiality, evidence-handling, and de-escalation procedures.

Technical Skills

Languages: Python, Java, SQL, JavaScript, HTML/CSS

Backend & APIs: FastAPI, REST APIs, Pydantic, Uvicorn/ASGI, Jinja2, HTTP/JSON, API integration

Machine Learning & Data: PyTorch, Hugging Face Transformers, Sentence Transformers, scikit-learn, NumPy, Pandas, NLP, text embeddings, cosine similarity, semantic ranking

Cloud & DevOps: AWS (ECS, ECR, IAM), Docker, Docker Compose, Linux, Git, GitHub

Databases & Tools: SQLite, relational database design, FFmpeg, yt-dlp, RapidFuzz